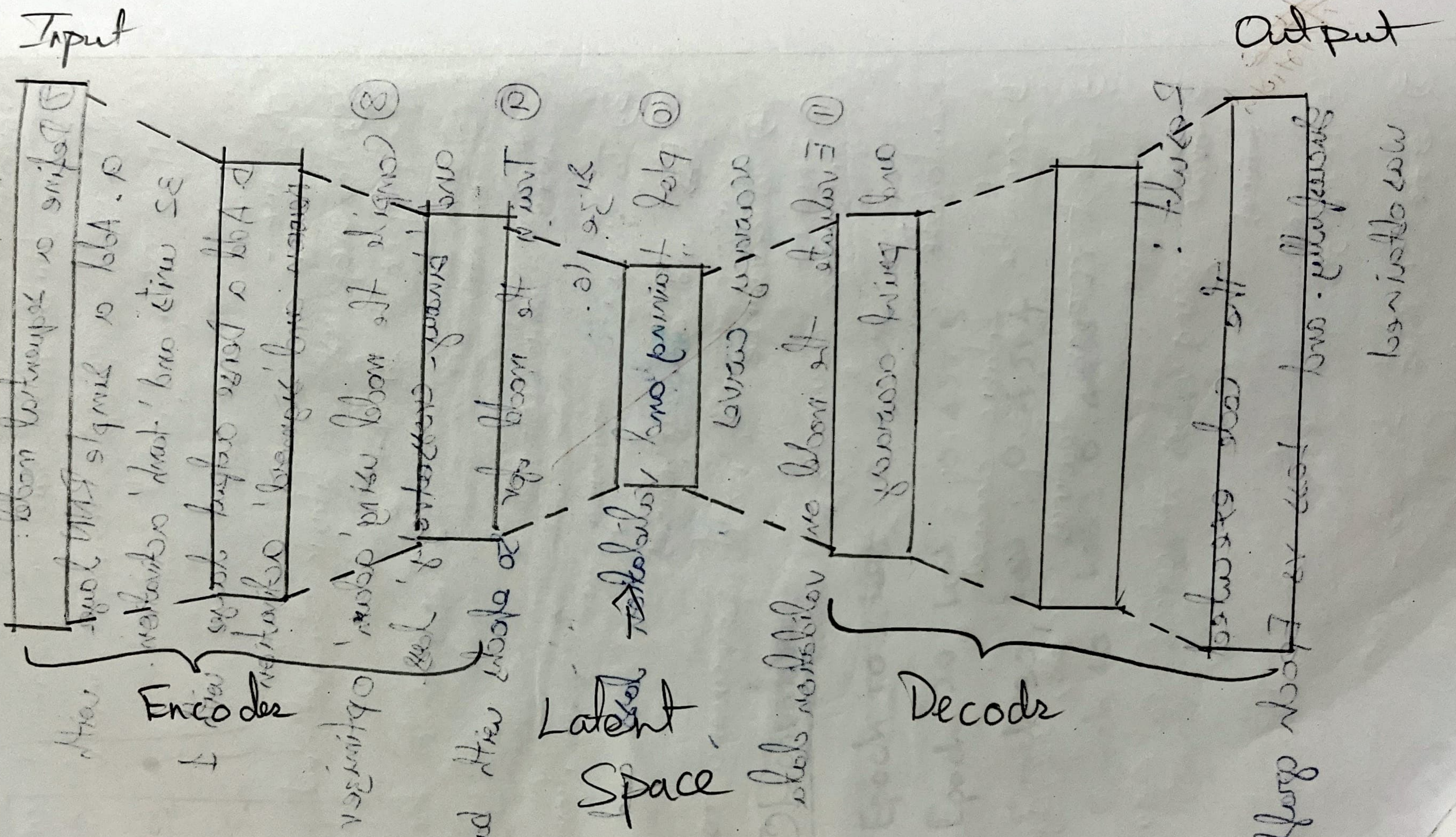

Input
Output
Encoder
Latent Space
Decoder

Exp.10

AutoEncoders

Aim:

To design and implement an autoencoder neural network that learns to compress and reconstruct handwritten digit images from the MNIST dataset, demonstrating the ability of autoencoders to perfor dimensionally reduction and data reconstion

Objective

- Understand the architecture of an autoencoder, including encoder and decoder parts
- Train the autoencoder on the MNIST dataset to minimize reconstruction loss.
- Train the model to minimize reconstruction loss between input and output
- Track and print epoch wise loss and accuracy during training
- Visualize traing and validation loss to evaluate model learning over epochs.

Pseudocode:-

① Load the dataset

② Preprocess the data
- select relevant features
- encode categorical variables

③ Flatten the images (28x28 -> 784)

④ Define the autoencoder architecture

encoder part
Input Layer : 784 neurons
hidden layer 1: 128 neurons, activation = ReLU
hidden layer 2: 64 neurons, activation = ReLU
Bottle neck : 32 neurons, activation = ReLU

Decoder part
Hidden layer 1: 64 neurons, activation = ReLU
Hidden layer 2, 128 neuron, activation = ReLU
Output layer = 784 neurons, activation = sigmoid

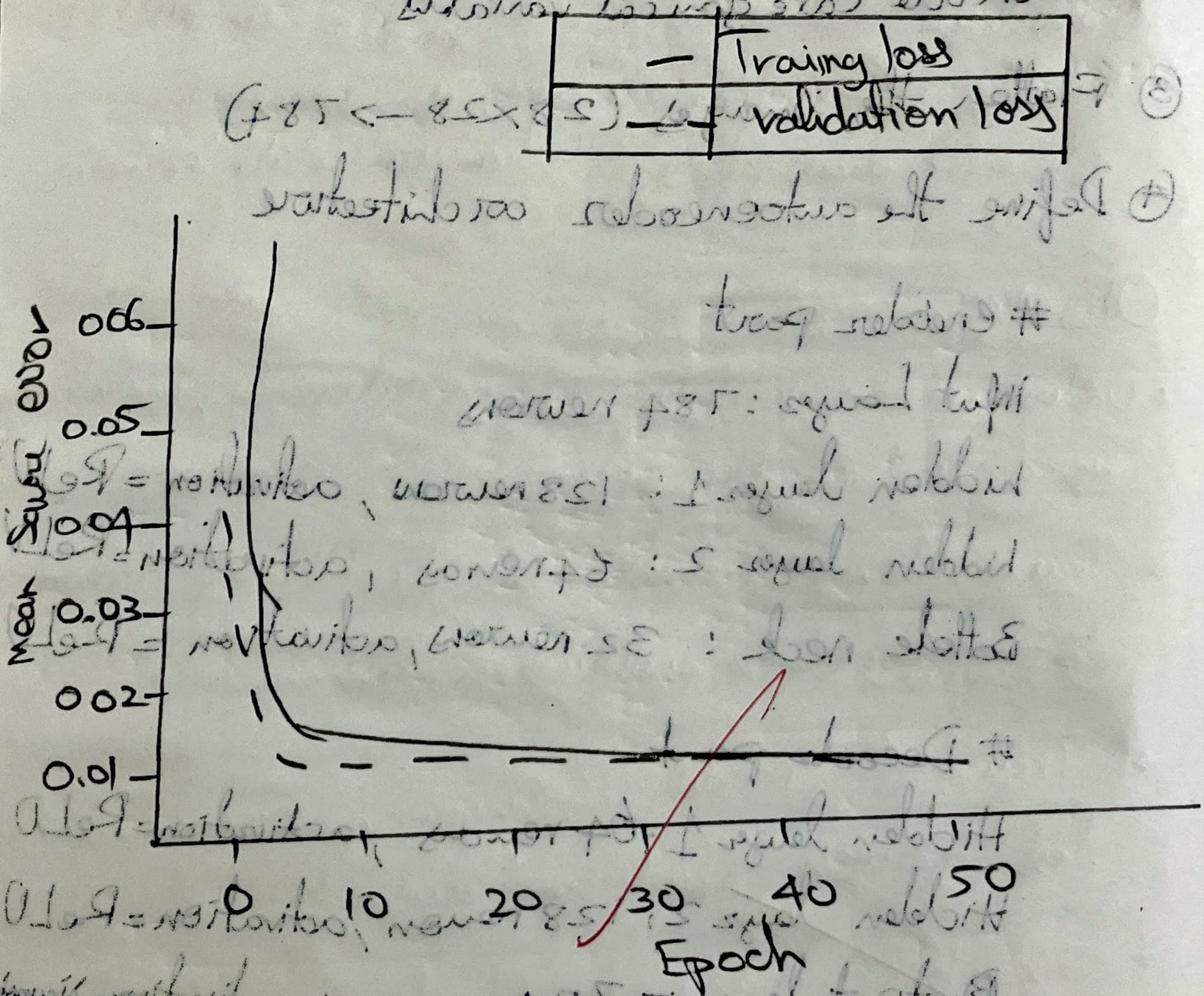
⑤ Connect encoder and decoder

⑥ compile the model

⑦ Train the autoencoder

Observation

Epoch 10 : Train loss : 0.013553 , Val loss = 0.013276
Epoch 20 : Train loss : 0.010141 , Val loss = 0.009889
Epoch 30 : Train loss : 0.008748 , Val loss = 0.008650
Epoch 40 : Train loss : 0.008040 , Val loss = 0.007860
Epoch 50 : Train loss : 0.007539 , Val loss = 0.007571



LOSS
VS
EPOCHS

8. extract the encoder model for compression
9. Perform compression
10. reconstruct the images
11. evaluate compression quality
12. Visualize results

Result

The code has executed successfully